

Experiment 6

Aim: Draw melting point-mass percent composition diagram for two component mixtures and determine the Eutectic Temperature.

List of Chemicals, Labwares and Equipment:

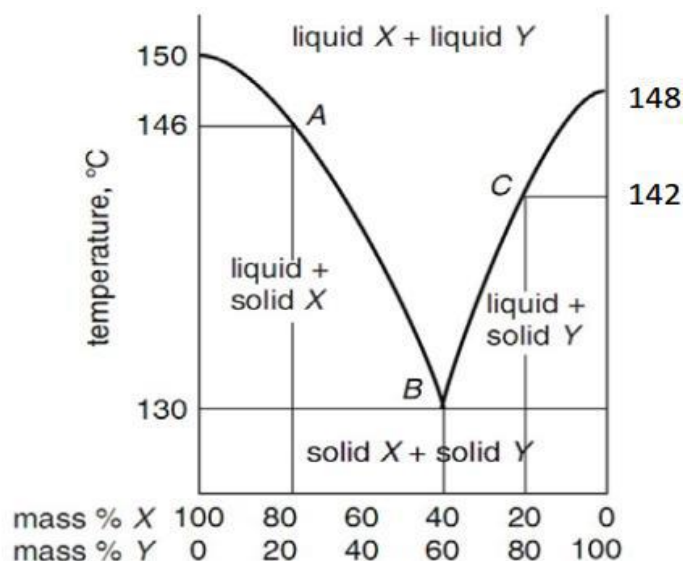
- a) Graph Paper, ruler scale (mm), butter paper (small pieces: 2 x2)
- b) Small spatula
- c) 2 Watch glasses
- d) 6-8 Melting point capillary tubes
- e) Trans-Cinnamic Acid: 50 mg [133-134 °C]
- f) Urea: 50 mg [133-134 °C]
- g) Melting Point Apparatus
- h) Analytical Balance

Theory

In the majority of cases the presence of a foreign substance will lower the melting point of a pure organic compound. This fact is utilized in the so-called mixed melting point test for the identification of organic compounds.

The melting point of a compound is the temperature at which the solid phase is in equilibrium with the liquid phase. A solid compound changes to a liquid when the molecules acquire enough energy to overcome the forces holding them together in an orderly crystalline lattice. For most organic compounds, these intermolecular forces are relatively weak. The melting point range is defined as the span of temperature from the point at which the crystals first begin to liquefy to the point at which the entire sample is liquid. Most pure organic compounds melt over a narrow temperature range of 1-2 °C. The presence of a soluble impurity almost always causes a decrease in the melting point expected for the pure compound and a broadening of the melting point range. In order to understand the effects of impurities on melting point behavior, consider the melting point-mass percent composition diagram for two different fictitious organic compounds, X and Y, shown in Figure 1. The vertical axis represents temperature and the horizontal axis represents varying mass percent compositions of X and Y.

Figure 1. Melting point-mass percent composition diagram



Cases may arise in which the melting point of certain mixtures are higher than the individual components, e.g., if an addition compound of higher melting point is formed or if the two compounds are completely soluble in the solid state forming solid solutions. Furthermore, for certain optical isomers, e.g., d- and Z-camphoroximes and for d and Z-borneol, there is no depression in the melting point, the freezing or melting points of all mixtures being the same as the pure components. It will be seen, therefore, that the mixed melting point test, although of great practical value, is not infallible and should accordingly be used with reasonable regard to these possibilities.

ii) Procedure

1. Determine the melting point of pure cinnamic acid (133°) and pure urea (133°).
2. Intimately mix equal weights (ca. 20 mg.) of the two finely-powdered compounds and determine the melting point; a considerable depression of melting point will be observed. It is instructive to construct a melting point diagram for mixtures of cinnamic acid and urea. Mix 80 mg of cinnamic acid (X) with 20 mg of urea (Y) intimately with the aid of a spatula on a glass slide, and determine the melting point (the temperature at which the mixture just becomes completely fluid is noted). Similarly prepare another three set of a mixed sample (in mg) for 60-40, 40-60, and 20-80. Plot temperatures as ordinates, and, concentration as abscissae, the percentage of urea from left to right (0-100 per cent.) and of cinnamic acid from right to left (0-100 per cent.).

3. Observation

Tabulate your results as follows:—

CINNAMIC ACID	UREA	M.P.
100 %	0 %	133°
80 %	20 %	—
60 %	40 %	—
50 %	50 %	—
40 %	60 %	—
20 %	80 %	—
0 %	100 %	133°

4. Results: Write the conclusion.

5. Precautions:

1. Mixed the samples homogeneously
2. Fill the capillary carefully without breaking the bottom
3. Observe carefully the complete melting of the mass inside the capillary during heating